🎓 MASSEY MACHINE — EXAM CHEAT SHEET

The whole course in one page: number systems, 2's complement, the instruction set, bitwise ops, shifts & the 5 program recipes.

Q1 = Massey · 6 marks | Instruction set provided | Exam-day companion

🔢 NUMBER SYSTEMS — THE SEAT PRICES

SYSTEM	SEATS (RIGHT → LEFT)	CARRY AT
Binary	`1, 2, 4, 8, 16, 32 …`	2
Hex	`1, 16, 256, 4096`	16

Letters: A=10 · B=11 · C=12 · D=13 · E=14 · F=15

- **Bin → Dec:** add seats that have a 1
- **Bin → Hex:** chop into 4s **from the RIGHT**, each group = 1 digit
- **Dec → Bin/Hex:** ÷ the base repeatedly, keep remainders, read **bottom → top**
- **Hex → Dec:** digit × seat price, add up
- **Hex → Bin:** each digit → its 4-bit twin

Famous values: `F`=15 · `FF`=255 · `FFFF`=65535 *(all-full register = one less than rollover)*

⚠️ **Trap:** "10" in hex = 16, NOT ten. Ten = `A`.

➖ NEGATIVES — 2'S COMPLEMENT

To NEGATE: flip partners + add 1 — works BOTH directions (to make AND to read).

Flip partners = 15 − digit:

```
0↔F   1↔E   2↔D   3↔C   4↔B
5↔A   6↔9   7↔8
```

- **Negate 5:** `0005` → flip → `FFFA` → +1 → `FFFB` ✓
- **Read `FFFD`:** flip → `0002` → +1 → 3 → it's **−3**

FIRST HEX DIGIT	SIGN
`0–7`	positive
`8–F`	negative

Why it works: `FFFB + 0005` → every column hits 16 → all zeros, carry falls off → `0000` ✓

Range: −32768 (`8000`) to +32767 (`7FFF`)

⚙️ THE INSTRUCTION SET

CODE	MEANING
`1xYZ`	LOAD Rx ← number YZ
`20xy`	LOAD Rx ← copy of Ry
`3xYZ`	LOAD Rx ← locker [YZ]
`4xYZ`	STORE Rx → locker [YZ]
`6xyz`	Rx = Ry + Rz
`70x0`	NEGATE Rx

CODE	MEANING
`8xYZ`	SHIFT RIGHT (÷2)
`9xYZ`	SHIFT LEFT (×2)
`Axyz`	Rx = Ry AND Rz
`Bxyz`	Rx = Ry OR Rz
`Cxyz`	Rx = Ry XOR Rz
`DxYZ`	JUMP to YZ **if Rx == R0**
`E000`	HALT

R0 = always 0 → the comparison buddy. `D00Z` = always-jump (R0==R0 is always true) — used to skip code / loop back.

🎭 BITWISE TRIO — COLUMN BY COLUMN, NO CARRIES

OP	RESULT IS 1 WHEN…
AND	**both** are 1
OR	**any** is 1
XOR	they are **different**

- `8000` AND x → isolates the **sign bit**: result 0000 = positive, 8000 = negative
- `0001` AND x → isolates the **last bit** (used in multiplication)

⚠️ **AND on hex:** convert to binary first, AND there, convert back.
Shortcut: first hex digit 8–F → `x AND 8000` = `8000` (negative); 0–7 → `0000`.

🔀 SHIFTS

- **LEFT = ×2** — zeros enter from the right, left bits **fall off**
- **RIGHT = ÷2** — bits fall off the right
- Register = fixed 16 bits = 4 hex digits — **the box never grows!**
- Shifting 8 bits = shifting 2 hex digits (`0080 → 8000`)

Why LOAD + SHIFT? LOAD only carries 2 hex digits (max FF) — so **load small, shift into place** to build `8000`.

📿 THE 5 PROGRAM RECIPES

1️⃣ Addition (Ex 1)

LOAD inputs → ADD → STORE → HALT

2️⃣ Subtraction (Ex 2)

No subtract instruction! LOAD both → **NEGATE** the second → ADD → STORE.

a − b = a + (−b)

3️⃣ If-Else (Ex 4)

No ">" instruction! → **subtract → AND with 8000 (mask) → jump if result == 0** (positive/true).

Layout: decision jump → **ELSE body first** → escape jump → **IF body** → store

4️⃣ Loop (Ex 5)

= if-else **+ one backwards jump**: test → body → jump back to test → repeat until test fails → exit

5️⃣ Multiplication (Ex 6)

No × instruction! R1 **doubles** each round, R2 **halves** each round; when R2's last bit = 1 → add R1 to total. Stop when R2 = 0.

3×5: 5 = 4+1 → add 3×1 and 3×4 = 3 + 12 = 15

🏆 How to attack Q1

Trace: build a table (PC, registers, memory), update line by line. **Write:** pick the recipe that matches the task, adapt addresses. **Convert:** scribble the seat prices first.

🐛 TOP TRAPS TO DODGE

1. Numbers in instructions are ALWAYS hex (decimal is just the human subtitle).
2. "10" = sixteen, ten = `A`.
3. Register = fixed 4 hex digits — overflow falls off (a feature, not a bug!).
4. A register never loaded = garbage — always trace from the top.
5. After STORE, the register keeps its value (it's a copy).
6. AND / OR / XOR = bitwise = per column, never carries.
7. When tracing: build a table — PC, each register, memory — update line by line.

✍️ EXAM-DAY SCRIBBLES (WRITE THESE FIRST!)

```
Seats: 4096 256 16 1  ·  8 4 2 1  ·  A10 B11 C12 D13 E14 F15
Flip:  0↔F 1↔E 2↔D 3↔C 4↔B 5↔A 6↔9 7↔8  (partner = 15 − digit)
Sign:  hex 0–7 = positive · 8–F = negative
```

From "what is a register?" to tracing a multiplication algorithm — you've got this. **Go smash Q1!** 💪